



2023 CSO discharges in the Lower Wye Catchment

Analysis of EDM data for the Wye below Monmouth

A report prepared on behalf of Wye Salmon Association (WSA)

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Introduction

This report attempts to summarise sewage discharge pressures in the Wye catchment downstream of Monmouth. It is a follow-up to an earlier report (Green, 2024) on the 2023 EDM data for the Wye MC.

Sewage pressures comprise two elements, the first of which is the normal operation of sewage treatment works (STWs). Even though the effluent from these is “secondary processed”, this clearly does not imply non-polluting. STW effluent contains significant loads of nutrients, organic matter, detergent residues, pharmaceuticals, harmful bacteria, pesticides, etc.

Secondly, we have discharges of unprocessed and partially processed sewage, exposed by the EDM statistics. Combined sewers, as constructed in earlier times, take rainwater as well as foul water and get overloaded in heavy rain. More modern infrastructure separates surface water and foul water, with only the latter being routed to the STW. But even these suffer from cross connections¹. Infiltration of groundwater can also be a problem on older systems. Combined sewer overflows (CSOs) are typically simple weir arrangements to automatically vent excess flow and prevent back-up. The EDM sensor simply detects the fact that liquid is flowing over the weir; it tells us nothing about flow volume. At the STW, as well as CSOs on the input, we also find storm tank overflows which discharge ‘partially settled’ effluent - so with less solid matter. Pumping stations are another special case. Discharges from pumping stations can often be a symptom of pump failure rather than rainfall.

Published EDM data is summarised on an annual basis. It tells us nothing about the timing of discharges. So, we have no idea about any deliberate, non-rainfall driven (hence illegal) discharging from STWs, as has been exposed at some of the water companies. This requires a more detailed, forensic analysis, such as those performed by Peter Hammond. Indeed, Peter has identified illegal discharging by [DCWW at many sites across Wales](#), including Bult Wells on the Wye.

STWs

There are 6 STWs in the (non-tidal) lower catchment, as listed in table 1. Locations are shown in figure 1 overleaf.

Name	Input load (PE) ²	Receiving water
Rockfield STW	34	Monnow
Dingestow STW	1,799	Trothy
Monmouth STW	11,777	Wye
Newland STW	17,098	Wye & Valley Brook
St Briavels STW	706	Wye
Llandogo STW	507	Wye

Table 1: sewage treatment works in the lower Wye catchment

Newland and Monmouth are the standouts in terms of input load, with the former having had P-stripping tertiary treatment for several years. Rockfield is tiny and, based on dilution ratio, wouldn't be expected to have much impact on the Monnow. Dingestow, on the other hand, is probably very significant to the smaller Trothy, particularly in summer, and is pencilled in for P-stripping in AMP-8. Of course, this all pales into insignificance against the combined load of 121,000 PE coming down from Hereford and Ross.

¹ Cross connections are troublesome in the other direction too. Post-build modifications to houses frequently put foul water into the surface water drains – which are routed directly to the rivers, not via the STW. And consider all the car-wash businesses operating on supermarket car parks, whose detergent-loaded effluent goes into surface water drains.

² PE, population equivalent is the standard unit of input load for STWs, representing the load of 1 average person.

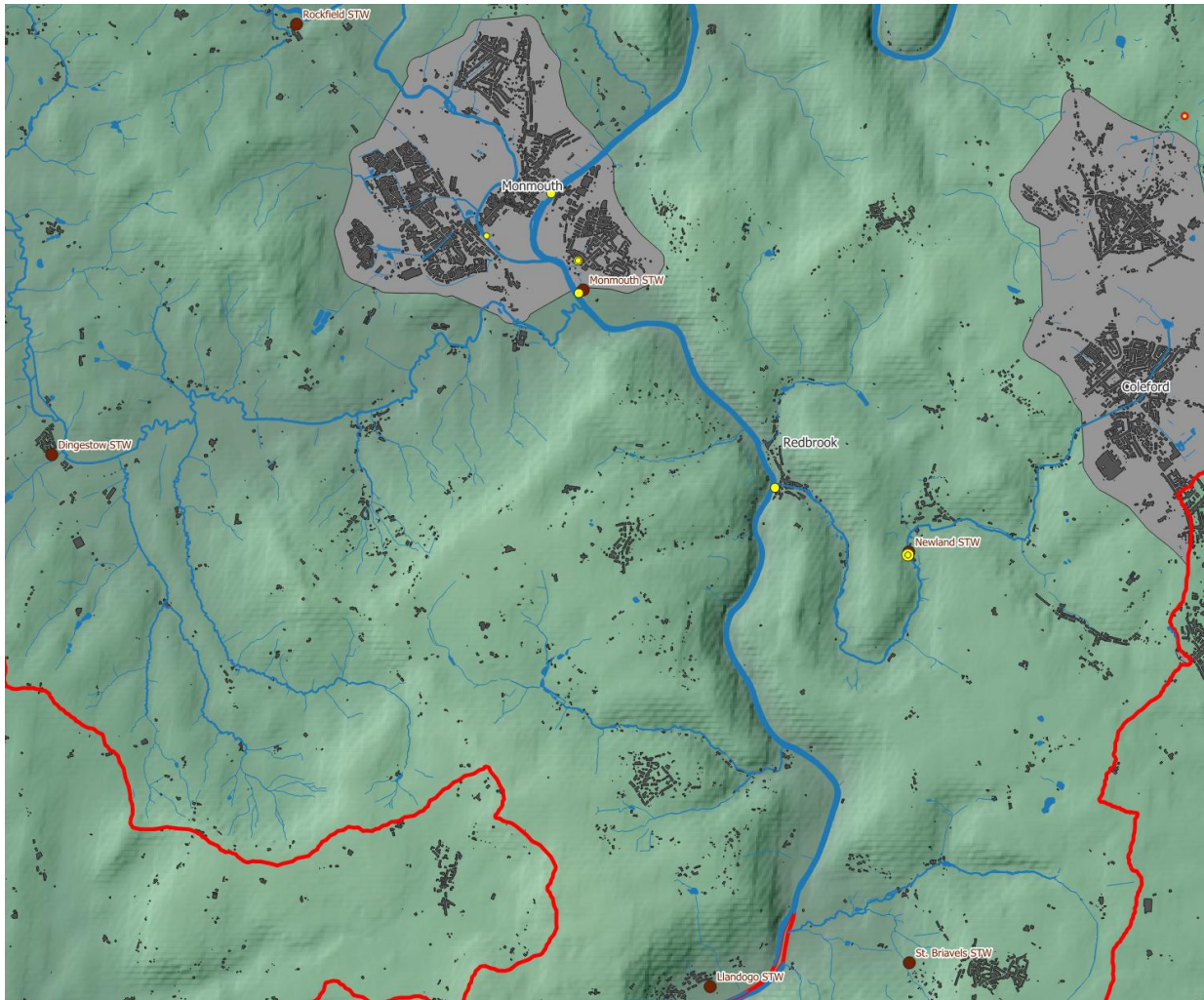


Figure 1: locations of STWs (brown) and 2023 CSO discharges (yellow, scaled to discharge duration)

The larger STWs frequently have multiple permits, so before diving into EDM data, we shall review the DCWW asset list provided to us in 2021 under an FoI request. The situation for Monmouth, Newland & St Briavels STWs is summarised in table 2 below.

Name	Permit no.	Discharge type	Receiving water
MONMOUTH STW	AN0353501	Settled Storm Sewage	River Wye
MONMOUTH STW	AS1003201	Secondary Treated Effluent	River Wye
NEWLAND STW	AN0294201	Storm Sewage	Valley Brook
NEWLAND STW	AN0294901	Settled Storm Sewage	Valley Brook
NEWLAND STW	AS1003301	Secondary Treated Effluent	River Wye
NEWLAND STW	AS1003301	Settled Storm Sewage	River Wye
ST BRIAVELS STW	AS1004501	Secondary Treated Effluent	R. Wye Estuary
ST BRIAVELS STW	AS1004501	Storm Sewage	R. Wye Estuary

Table 2: license details for the larger STWs

Monmouth STW has a license to discharge treated effluent, around 2,000 m³/day, into the Wye. It also has a separate licence to discharge “settled” sewage from the same location during heavy rainfall. Newland has two analogous permits to discharge, via pipework, directly into the Wye at Redbrook, in this case about 3,000 m³/day in normal operation. But note that Newland also has two permits to discharge both settled and raw sewage into the Valley Brook. Confusingly, from 2022-23, some of these outfalls appear to have been amalgamated into the AS1003301 license. So, some care is needed in

comparing year to year EDM data. St Briavels has a single license covering both normal operation (about 130 m³/day) and storm sewage.

EDM data

EDM data falling within the lower catchment is centred around Monmouth, Newlands/Redbrook and St Briavels. Table 3 summarises the data for 2020 and 2023 (recall that 2020 & 2023 were similar, high-rainfall years).

		2020		2023	
License Ref	Discharge name	Discharge events	Duration (hr)	Discharge events	Duration (hr)
Monmouth town:					
AN0083602	CSO at Monmouth Town Pumping Station	23	115	25	38
AN0153301	Monmouth 2 CSO	-	-	58	108
Unpermitted-72709	4 Hillcrest Road CSO, Monmouth	-	-	81	149
Unpermitted-72710	24 Hillcrest Road CSO, Monmouth	-	-	0	0
Monmouth STW:					
AN0353501	Settled storm sewage from Monmouth WwTW	86	1163	66	545
AN1003201	Monmouth Wastewater Treatment Works	-	-	24	263
Newland STW into R. Wye at Redbrook:					
AN294101	Newland STW	68	863	-	-
AS1003301	Newland STW (storm tank A2)	-	-	56	603
Newland STW into Valley Brook:					
AN294201	Newland STW	25	89	-	-
AN294901	Newland STW	82	1291	-	-
AS1003301	CSO at Newland STW (storm tank A3)	-	-	23	32
AS1003301	CSO at Newland STW (storm tank A4)	-	-	85	1675
St Briavels STW into tidal Wye:					
AS1004501	St Briavels STW	29	97	-	-

Table 3: EDM data for untreated sewage discharges in the lower Wye catchment

Of particular note is the performance of the STWs at Monmouth and Newland. At Monmouth, we see a combined discharge duration of just over a month in 2023. The 2020 number is broadly similar.

At Newland STW, we find storm tank “A2” discharging for a similar length of time into the Wye at Redbrook. Assuming a correspondence to the 2020 “AN294101” license, again, nothing has improved in the intervening years. The Valley Brook discharges are even higher, with 10 weeks of discharge during 2023 from “A4”. This effluent also ends up in the Wye at Redbrook, after passing through the Bigwell Fly Fishery ponds en route (I certainly wouldn’t be eating my catch from there). As above, there is an obvious correspondence to the 2020 licenses “AN294201” & “AN294901”, suggesting that nothing has changed.

St Briavels STW is less significant, having a relatively small output in normal operation and discharging into the tidal river. In 2021 it recorded 61 spills with a total duration of 587 hours – much higher than in 2020. Since then, it has not featured in the EDM data. This suggests that perhaps there was a fault condition which has been rectified.

Averaging the total discharge durations for this selection of sites yields the graph of figure 2 overleaf. This is very similar to the equivalent graph for the whole of the Wye MC (figure 5 in Green, 2024) and is thus indicative of being primarily rainfall driven. Moreover, the values of duration per site are similar, meaning that DCWW performance in the lower catchment is typical of the wider Wye catchment.

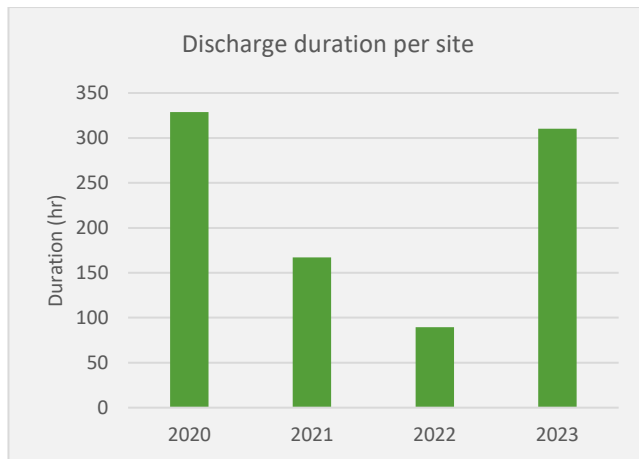


Figure 2: average discharge duration per site within the lower Wye catchment (data from table 3)

Conclusion

- The EDM data for the lower Wye catchment is typical of the full Wye MC data, showing a similar rainfall-dominated year to year trend.
- In common with the Wye MC picture, the EDM data for the lower catchment show little evidence of improvement between 2020 and 2023.
- Storm tank emissions from Monmouth STW and Newland STW are worrying. But in the absence of volumetric data, it is impossible to know the ecological impact.
- Even in normal operation, Monmouth and Newland undoubtedly represent substantial local pollution loads.

References

Green, G. (2024). *2023 CSO discharges in the Wye Catchment*.